

ECE 3317 Final Exam

Step-by-Step Problem Solving Guide

Master These Procedures = Ace the Exam

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1 Quarter-Wave Transformer Matching

1.1 Problem Type Recognition

You're given:

- A load impedance Z_L (often complex like $25 - j50 \Omega$)
- A main feed line with characteristic impedance Z_0 (often 50Ω)
- Asked to find distance d to place transformer and transformer impedance Z_{0T}

1.2 Complete Solution Procedure

Step 1: Calculate the normalized load impedance

$$Z_L^N = \frac{Z_L}{Z_0}$$

Example: If $Z_L = 25 - j50 \Omega$ and $Z_0 = 50 \Omega$, then $Z_L^N = 0.5 - j1.0$

Step 2: Plot Z_L^N on the Smith chart

- Real part: find the resistance circle (0.5 in example)
- Imaginary part: find the reactance arc (-1.0 means lower half of chart)
- Mark the intersection point

Step 3: Read the “wavelengths toward generator” scale at Z_L^N

- Draw a line from center through your point to the outer scale
- Record this value (call it WTG_1)

Step 4: Draw constant- $|\Gamma|$ circle through Z_L^N

- Use a compass centered at the chart center
- Circle passes through your load point

Step 5: Find where circle crosses the REAL AXIS (positive side)

- Quarter-wave transformer needs a **real** input impedance
- Read the normalized resistance value R_{in}^N at this crossing
- Read WTG_2 at this point

Step 6: Calculate distance d

$$d = (\text{WTG}_2 - \text{WTG}_1) \times \lambda_g$$

- If result is negative, add $0.5\lambda_g$
- Problem usually asks for “smallest d ” — use the first crossing

Step 7: Calculate the unnormalized input impedance

$$Z_{in} = R_{in}^N \times Z_0$$

Step 8: Calculate transformer characteristic impedance

$$Z_{0T} = \sqrt{Z_0 \times Z_{in}}$$

Step 9: Calculate physical length of transformer (if asked)

$$\lambda_g = \frac{c}{f\sqrt{\epsilon_r^{eff}}}$$

$$l_T = \frac{\lambda_g}{4}$$

WARNING: The transformer input impedance **MUST** be real. If you're not on the real axis, you made an error.

TIP: If $Z_{in} < Z_0$, then $Z_{0T} < Z_0$. If $Z_{in} > Z_0$, then $Z_{0T} > Z_0$. Use this to sanity-check.

1.3 General Transmission Line Input Impedance

For ANY transmission line of length l with load Z_L :

$$Z_{in} = Z_0 \frac{Z_L + jZ_0 \tan(\beta l)}{Z_0 + jZ_L \tan(\beta l)}$$

where $\beta = 2\pi/\lambda_g$ is the phase constant.

Special Cases (MEMORIZE THESE):

Line Length	Input Impedance	Notes
$l = \lambda/4$	$Z_{in} = Z_0^2/Z_L$	Impedance inverter
$l = \lambda/2$	$Z_{in} = Z_L$	Impedance repeater
$l = \lambda$	$Z_{in} = Z_L$	Same as $\lambda/2$
Short circuit ($Z_L = 0$)	$Z_{in} = jZ_0 \tan(\beta l)$	Pure reactance
Open circuit ($Z_L = \infty$)	$Z_{in} = -jZ_0 \cot(\beta l)$	Pure reactance

1.4 Return Loss and Mismatch Loss

Return Loss (RL): How much power is reflected back

$$RL = -20 \log_{10} |\Gamma| \quad [\text{dB}]$$

Perfect match: $RL = \infty$ dB. Larger RL = better match.

Mismatch Loss (ML): How much power is NOT delivered to load

$$ML = -10 \log_{10}(1 - |\Gamma|^2) \quad [\text{dB}]$$

Relationship between SWR and $|\Gamma|$:

$$|\Gamma| = \frac{SWR - 1}{SWR + 1}$$

2 Single Stub Matching

2.1 Problem Type Recognition

You're given:

- A load impedance Z_L
- Main line and stub both have characteristic impedance Z_0
- Asked to find distance d from load to stub, and stub length l_s
- Stub is either open-circuited or short-circuited

2.2 Complete Solution Procedure

Step 1: Normalize the load impedance

$$Z_L^N = \frac{Z_L}{Z_0}$$

Step 2: Plot Z_L^N on Smith Chart #1

Step 3: Convert to normalized admittance Y_L^N

- Rotate 180° around the chart (reflect through center)
- OR: $Y_L^N = 1/Z_L^N$
- Mark this point and record its WTG value (WTG₁)

Step 4: Draw constant- $|\Gamma|$ circle through Y_L^N

Step 5: Find where circle crosses the $g = 1$ circle

- The $g = 1$ circle passes through the center of the chart
- There are TWO intersections — choose the one requiring smaller d
- At intersection, read: $Y_{in}^N = 1 + jb$
- Record WTG₂ at this point

Step 6: Calculate distance d

$$d = (\text{WTG}_2 - \text{WTG}_1) \times \lambda_g$$

Add $0.5\lambda_g$ if negative.

Step 7: Determine required stub susceptance

$$b_{stub} = -b$$

(Must cancel the susceptance at the attachment point)

Step 8: Find stub length using Smith Chart #2

For OPEN-CIRCUITED stub:

- Start at $Y = 0$ (right side of chart, open circuit)
- WTG at start = 0.25
- Move toward generator (clockwise) until you reach $Y^N = jb_{stub}$
- Read WTG_{end}
- $l_s = (\text{WTG}_{end} - 0.25) \times \lambda_g$

For SHORT-CIRCUITED stub:

- Start at $Y = \infty$ (left side of chart, short circuit)
- WTG at start = 0
- Move toward generator until you reach $Y^N = jb_{stub}$
- $l_s = \text{WTG}_{end} \times \lambda_g$

WARNING: Stubs are in PARALLEL, so you MUST work with ADMITTANCES, not impedances!

WARNING: For double stub matching with two identical stubs: each stub provides half the required susceptance. So $b_{each} = -b/2$.

TIP: The $g = 1$ circle is the vertical line passing through the center of the Smith chart.

2.3 Stub Input Impedance Formulas (Without Smith Chart)

Short-Circuited Stub:

$$\boxed{Z_{in}^{SC} = jZ_0 \tan(\beta l_s)} \quad Y_{in}^{SC} = \frac{-j}{Z_0} \cot(\beta l_s)$$

Open-Circuited Stub:

$$\boxed{Z_{in}^{OC} = -jZ_0 \cot(\beta l_s)} \quad Y_{in}^{OC} = \frac{j}{Z_0} \tan(\beta l_s)$$

2.4 Lumped Element Equivalents

Stubs can be replaced by lumped capacitors or inductors:

Stub Type	Short Stub ($l < \lambda/4$)	Equivalent
Short-circuited	$Z_{in} = jZ_0 \tan(\beta l) > 0$	Inductor: $L = \frac{Z_0 \tan(\beta l)}{\omega}$
Open-circuited	$Z_{in} = -jZ_0 \cot(\beta l) < 0$	Capacitor: $C = \frac{\tan(\beta l)}{\omega Z_0}$

If given a shunt capacitor or inductor instead of stub:

- Shunt capacitor: $Y_C = j\omega C$ (positive susceptance)
- Shunt inductor: $Y_L = -j/(\omega L) = -j/(X_L)$ (negative susceptance)

3 Smith Chart at Non-Design Frequency

3.1 Problem Type Recognition

You designed a matching network at frequency f_1 , but now asked about behavior at frequency f_2 .

3.2 Complete Solution Procedure

Step 1: Determine electrical length scaling

At the original design frequency, you moved a certain number of wavelengths. At the new frequency:

$$\text{New electrical length} = \text{Original electrical length} \times \frac{f_2}{f_1}$$

Example: Quarter-wave at f_1 becomes $(1/4)(f_2/f_1)$ wavelengths at f_2

Step 2: Recalculate guided wavelength if needed

$$\lambda_{g,new} = \frac{c}{f_2 \sqrt{\epsilon_r^{eff}}}$$

Step 3: Plot the load and rotate by new electrical length

Step 4: Read the new input impedance

Step 5: Renormalize if connecting to different Z_0 line

$$Z_{in}^{N,new} = \frac{Z_{in}(\text{unnormalized})}{\text{new } Z_0}$$

Step 6: Find SWR

- Draw constant- $|\Gamma|$ circle through Z_{in}^N
- Read where it crosses positive real axis
- That value IS the SWR

TIP: At double the frequency, a $\lambda/4$ section becomes $\lambda/2$, which means the impedance transforms back to itself!

4 Rectangular Waveguide Design

4.1 Problem Type A: Design Dimensions

Given: Operating frequency f_0 should be at center of single-mode band.

Step 1: Recognize the design condition

- Single-mode band: only TE_{10} propagates
- Band is between $f_{c,TE_{10}}$ and $f_{c,TE_{20}}$
- Center frequency: $f_0 = \frac{1}{2}(f_{c,10} + f_{c,20}) = 1.5 \times f_{c,10}$

Step 2: Solve for dimension a

$$f_0 = 1.5 \times \frac{c}{2a}$$

$$a = \frac{1.5c}{2f_0} = \frac{0.75c}{f_0}$$

Step 3: Determine dimension b

For maximum power without sacrificing bandwidth:

$$b = \frac{a}{2}$$

This makes $f_{c,TE_{01}} = f_{c,TE_{20}}$.

Step 4: Verify (optional but good practice)

- $f_{c,10} = c/(2a)$
- $f_{c,20} = c/a$
- Check that $f_0 = 1.5 \times f_{c,10}$ (verify!)

4.2 Problem Type B: Attenuation Below Cutoff

Given: Waveguide dimensions a , b and frequency $f < f_c$. Find attenuation in dB.

Step 1: Calculate the cutoff frequency

$$f_{c,10} = \frac{c}{2a}$$

Step 2: Verify $f < f_c$ (otherwise wave propagates, no attenuation formula)

Step 3: Calculate k_0

$$k_0 = \frac{2\pi f}{c} = \frac{2\pi}{\lambda_0}$$

Step 4: Calculate attenuation constant α

$$\alpha = \sqrt{\left(\frac{\pi}{a}\right)^2 - k_0^2}$$

Units: Np/m (nepers per meter)

Step 5: Convert to dB

$$\text{Attenuation (dB)} = 8.686 \times \alpha \times z$$

where z is the distance traveled in meters.

WARNING: Make sure to convert a to meters before calculating! Common error: using cm.

WARNING: Below cutoff, the wave is evanescent. The attenuation is HUGE (often 100+ dB for short distances).

TIP: Memorize: 8.686 converts Np to dB. Comes from $20/\ln(10) = 8.686$.

4.3 Problem Type C: Power Capacity

Step 1: Calculate β at operating frequency

$$\beta = k_0 \sqrt{1 - \left(\frac{f_c}{f}\right)^2}$$

Step 2: Apply power formula

$$P_z = \frac{ab\beta}{4\omega\mu_0} |E_{max}|^2$$

where $|E_{max}| = E_c = 3 \times 10^6$ V/m for air breakdown.

Step 3: Calculate $\omega = 2\pi f$ and plug in

4.4 Phase Velocity, Group Velocity, and Guide Wavelength

Guide Wavelength (wavelength inside waveguide):

$$\lambda_g = \frac{\lambda_0}{\sqrt{1 - (f_c/f)^2}} = \frac{\lambda_0}{\sqrt{1 - (\lambda_0/\lambda_c)^2}}$$

Note: $\lambda_g > \lambda_0$ always (wave is “stretched” in guide).

Phase Velocity (speed of phase fronts):

$$v_p = \frac{\omega}{\beta} = \frac{c}{\sqrt{1 - (f_c/f)^2}}$$

Note: $v_p > c$ (superluminal, but no information travels faster than c).

Group Velocity (speed of energy/information):

$$v_g = \frac{d\omega}{d\beta} = c\sqrt{1 - (f_c/f)^2}$$

Note: $v_g < c$ always.

Important Relationship:

$$v_p \cdot v_g = c^2$$

4.5 Wave Impedance in Waveguide

TE Modes:

$$Z_{TE} = \frac{\eta_0}{\sqrt{1 - (f_c/f)^2}} = \frac{377}{\sqrt{1 - (f_c/f)^2}} [\Omega]$$

Note: $Z_{TE} > 377 \Omega$ (higher than free space).

TM Modes:

$$Z_{TM} = \eta_0 \sqrt{1 - (f_c/f)^2} = 377 \sqrt{1 - (f_c/f)^2} [\Omega]$$

Note: $Z_{TM} < 377 \Omega$ (lower than free space).

4.6 Dielectric-Filled Waveguide

If waveguide is filled with dielectric ($\epsilon_r, \mu_r = 1$):

$$f_c^{filled} = \frac{f_c^{air}}{\sqrt{\epsilon_r}} \quad \lambda_c^{filled} = \lambda_c^{air}$$

The cutoff frequency DECREASES, but the cutoff wavelength stays the same (it's set by geometry).

4.7 Which Modes Can Exist?

TE_{mn} modes: m or n can be zero (but not both). TE₁₀ is dominant.

TM_{mn} modes: Neither m nor n can be zero. Lowest is TM₁₁.

For standard waveguide ($b = a/2$):

$$f_{c,TM_{11}} = \frac{c}{2} \sqrt{\frac{1}{a^2} + \frac{1}{b^2}} = \frac{c}{2} \sqrt{\frac{1}{a^2} + \frac{4}{a^2}} = \frac{c\sqrt{5}}{2a}$$

5 Plane Wave Reflection (TM and TE)

5.1 Problem Type Recognition

Wave incident on interface between two media. Asked for:

- Reflection coefficients Γ_{TM} and Γ_{TE}
- Percentage of power reflected
- Effect of polarizing filter

5.2 Complete Solution Procedure

Step 1: Identify the media properties

- Medium 1 (incident): ε_{r1} , $n_1 = \sqrt{\varepsilon_{r1}}$
- Medium 2 (transmitted): ε_{r2} , $n_2 = \sqrt{\varepsilon_{r2}}$

Step 2: Apply Snell's Law to find θ_t

$$\boxed{n_1 \sin \theta_i = n_2 \sin \theta_t}$$

$$\theta_t = \sin^{-1} \left(\frac{n_1}{n_2} \sin \theta_i \right)$$

WARNING: If $\frac{n_1}{n_2} \sin \theta_i > 1$, you have TOTAL INTERNAL REFLECTION. Skip to that section.

Step 3: Calculate wave impedances

TM polarization:

$$Z_1^{TM} = \frac{\eta_0}{\sqrt{\varepsilon_{r1}}} \cos \theta_i = \frac{377}{\sqrt{\varepsilon_{r1}}} \cos \theta_i$$

$$Z_2^{TM} = \frac{\eta_0}{\sqrt{\varepsilon_{r2}}} \cos \theta_t = \frac{377}{\sqrt{\varepsilon_{r2}}} \cos \theta_t$$

TE polarization:

$$Z_1^{TE} = \frac{\eta_0}{\sqrt{\varepsilon_{r1}} \cos \theta_i} = \frac{377}{\sqrt{\varepsilon_{r1}} \cos \theta_i}$$

$$Z_2^{TE} = \frac{\eta_0}{\sqrt{\varepsilon_{r2}} \cos \theta_t} = \frac{377}{\sqrt{\varepsilon_{r2}} \cos \theta_t}$$

Step 4: Calculate reflection coefficients

$$\boxed{\Gamma_{TM} = \frac{Z_2^{TM} - Z_1^{TM}}{Z_2^{TM} + Z_1^{TM}}}$$

$$\boxed{\Gamma_{TE} = \frac{Z_2^{TE} - Z_1^{TE}}{Z_2^{TE} + Z_1^{TE}}}$$

Step 5: Calculate percentage of power reflected

If single polarization:

$$\%P_r = |\Gamma|^2 \times 100$$

If randomly polarized (equal TM and TE):

$$\%P_r = \left(\frac{1}{2} |\Gamma_{TM}|^2 + \frac{1}{2} |\Gamma_{TE}|^2 \right) \times 100$$

Step 6: If polarizing filter is involved

Filter blocks TE, passes TM:

$$\%P_{through} = \frac{1}{2} |\Gamma_{TM}|^2 \times 100$$

(The $\frac{1}{2}$ accounts for only half the incident power being TM)

TIP: Memory trick for impedance formulas:

- TM: E-field has component parallel to plane $\rightarrow Z^{TM} = \eta \cos \theta$ (cosine on top)
- TE: E-field perpendicular to plane $\rightarrow Z^{TE} = \eta / \cos \theta$ (cosine on bottom)

5.3 Transmission Coefficients

Power must be conserved: $P_{incident} = P_{reflected} + P_{transmitted}$

Field Transmission Coefficient:

$$T = 1 + \Gamma = \frac{2Z_2}{Z_2 + Z_1}$$

Power Transmission:

$$\frac{P_{trans}}{P_{inc}} = 1 - |\Gamma|^2 = \frac{Z_1}{Z_2} |T|^2 \frac{\cos \theta_t}{\cos \theta_i}$$

For normal incidence ($\theta_i = \theta_t = 0$):

$$\frac{P_{trans}}{P_{inc}} = \frac{4n_1 n_2}{(n_1 + n_2)^2}$$

5.4 Normal Incidence ($\theta_i = 0$)

When wave hits interface straight-on, the formulas simplify:

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{n_1 - n_2}{n_1 + n_2}$$

$$T = \frac{2\eta_2}{\eta_2 + \eta_1} = \frac{2n_1}{n_1 + n_2}$$

TM and TE give the SAME result at normal incidence (no distinction).

5.5 Quarter-Wave Anti-Reflection Coating

To eliminate reflection at normal incidence between media with n_1 and n_3 :

Coating Requirements:

- Thickness: $d = \lambda_2/4$ (quarter-wave in the coating material)
- Refractive index: $n_2 = \sqrt{n_1 \cdot n_3}$

The coating wavelength is: $\lambda_2 = \lambda_0/n_2$

5.6 Multi-Layer / Slab Problems

For a slab of thickness d between two media:

Step 1: Find transmission coefficient at first interface: T_{12} , Γ_{12}

Step 2: Propagate through slab: phase shift $e^{-jk_2 d \cos \theta_2}$

Step 3: Find transmission coefficient at second interface: T_{23} , Γ_{23}

Step 4: Account for multiple internal reflections (usually negligible if asked for first-order)

6 Total Internal Reflection & Evanescent Waves

6.1 Problem Type Recognition

- Wave traveling from denser to less dense medium ($n_1 > n_2$)
- Angle of incidence greater than critical angle
- Asked for attenuation in dB at some distance from surface

6.2 Complete Solution Procedure

Step 1: Calculate critical angle

$$\theta_c = \sin^{-1} \left(\frac{n_2}{n_1} \right)$$

Step 2: Verify total internal reflection

If $\theta_i > \theta_c$: TIR occurs, $|\Gamma_{TM}| = |\Gamma_{TE}| = 1$.

Step 3: Calculate k_0

$$k_0 = \frac{2\pi}{\lambda_0} = \frac{2\pi f}{c}$$

Step 4: Calculate evanescent wave attenuation constant

$$\alpha_{zt} = k_0 \sqrt{n_1^2 \sin^2 \theta_i - n_2^2}$$

Step 5: Calculate attenuation in dB at distance d

$$\text{Attenuation (dB)} = 8.686 \times \alpha_{zt} \times d$$

WARNING: The evanescent wave exists in medium 2 (the less dense medium). The distance d is measured perpendicular to the interface INTO medium 2.

TIP: This attenuation is typically VERY large. Even a fraction of a wavelength gives significant dB loss.

6.3 Axial Ratio of Reflected Wave

If incident wave is circularly polarized (RHCP or LHCP) and both Γ_{TM} and Γ_{TE} are real:

$$AR = \frac{|\Gamma_{TE}|}{|\Gamma_{TM}|} \quad (\text{assuming } |\Gamma_{TE}| > |\Gamma_{TM}|)$$

6.4 Fiber Optics: Numerical Aperture

For light to be guided in a fiber (total internal reflection), it must enter within the **acceptance cone**.

Critical Angle Inside Fiber:

$$\theta_c = \sin^{-1} \left(\frac{n_{clad}}{n_{core}} \right)$$

Numerical Aperture (NA):

$$NA = \sin \theta_{accept} = \sqrt{n_{core}^2 - n_{clad}^2}$$

Acceptance Angle (maximum angle from fiber axis for light to be guided):

$$\theta_{accept} = \sin^{-1}(NA)$$

Relationship: Light entering at angle θ_{accept} from air hits the core-cladding interface at exactly the critical angle.

TIP: For step-index fiber: if $n_{core} = 1.5$ and $n_{clad} = 1.48$, then $NA = \sqrt{1.5^2 - 1.48^2} = 0.244$, giving $\theta_{accept} = 14.1$.

6.5 Phase Shift Upon Total Internal Reflection

When TIR occurs, the reflection coefficients become complex with magnitude 1:

$$\Gamma_{TE} = e^{j\phi_{TE}}, \quad \Gamma_{TM} = e^{j\phi_{TM}}$$

The phase shifts are:

$$\tan\left(\frac{\phi_{TE}}{2}\right) = \frac{\sqrt{\sin^2 \theta_i - (n_2/n_1)^2}}{\cos \theta_i}$$

$$\tan\left(\frac{\phi_{TM}}{2}\right) = \frac{(n_1/n_2)^2 \sqrt{\sin^2 \theta_i - (n_2/n_1)^2}}{\cos \theta_i}$$

7 Lossy Medium Propagation

7.1 Problem Type Recognition

Wave traveling in conductive medium (like ocean water). Asked for attenuation in dB.

7.2 Complete Solution Procedure

Step 1: Identify medium parameters

- Relative permittivity ϵ_r
- Conductivity σ [S/m]
- Frequency f

Step 2: Calculate complex wavenumber

$$k = k_0 \sqrt{\epsilon_r - j \frac{\sigma}{\omega \epsilon_0}}$$

where $\omega = 2\pi f$ and $k_0 = \omega/c$.

Step 3: Extract imaginary part

Write $k = k' - jk''$ where k'' is the attenuation constant.

For a good conductor ($\sigma \gg \omega\epsilon$):

$$k'' \approx \sqrt{\frac{\omega \mu_0 \sigma}{2}} = \frac{1}{\delta}$$

where δ is skin depth.

Step 4: Calculate attenuation in dB

$$\text{Attenuation (dB)} = 8.686 \times k'' \times z$$

7.3 Skin Depth

The **skin depth** δ is the distance at which fields decay to $1/e$ (37%) of surface value:

General Formula:

$$\delta = \frac{1}{\alpha} = \frac{1}{k''}$$

For Good Conductor ($\sigma \gg \omega\epsilon$):

$$\delta = \sqrt{\frac{2}{\omega \mu \sigma}} = \sqrt{\frac{1}{\pi f \mu \sigma}}$$

Example values at 1 GHz:

- Copper ($\sigma = 5.8 \times 10^7$ S/m): $\delta \approx 2.1 \mu\text{m}$
- Aluminum ($\sigma = 3.5 \times 10^7$ S/m): $\delta \approx 2.7 \mu\text{m}$
- Seawater ($\sigma = 4$ S/m): $\delta \approx 0.25$ m

7.4 Loss Tangent

The **loss tangent** characterizes how “lossy” a material is:

$$\tan \delta_e = \frac{\sigma}{\omega \varepsilon} = \frac{\varepsilon''}{\varepsilon'}$$

Classification:

Condition	Material Type
$\tan \delta_e \ll 1$ (or $\sigma \ll \omega \varepsilon$)	Good dielectric (low loss)
$\tan \delta_e \gg 1$ (or $\sigma \gg \omega \varepsilon$)	Good conductor (high loss)
$\tan \delta_e \approx 1$	Neither (must use full formula)

7.5 Complex Permittivity

Lossy materials are described by complex permittivity:

$$\varepsilon_c = \varepsilon' - j\varepsilon'' = \varepsilon(1 - j \tan \delta_e) = \varepsilon - j \frac{\sigma}{\omega}$$

The wave equation uses:

$$k = \omega \sqrt{\mu \varepsilon_c} = k_0 \sqrt{\varepsilon_r - j \frac{\sigma}{\omega \varepsilon_0}}$$

7.6 Good Dielectric Approximation

When $\sigma \ll \omega \varepsilon$ (low loss):

$$\alpha \approx \frac{\sigma}{2} \sqrt{\frac{\mu}{\varepsilon}} = \frac{\sigma \eta}{2}$$

$$\beta \approx \omega \sqrt{\mu \varepsilon}$$

7.7 Good Conductor Approximation

When $\sigma \gg \omega \varepsilon$ (high loss):

$$\alpha = \beta = \sqrt{\frac{\omega \mu \sigma}{2}} = \frac{1}{\delta}$$

$$\eta_c = (1 + j) \sqrt{\frac{\omega \mu}{2\sigma}} = \frac{1 + j}{\sigma \delta}$$

8 Antenna Link Budget (Friis Equation)

8.1 Problem Type Recognition

- Satellite/CubeSat communicating with ground station
- Given: transmit power, antenna gains, distance, frequency
- Asked: power density at receiver, received power, Thévenin voltage

8.2 Complete Solution Procedure

Step 1: Convert all units to SI

- Distance: km $\rightarrow$ m (multiply by 1000)
- Frequency: GHz $\rightarrow$ Hz (multiply by 10^9)
- Gain in dB $\rightarrow$ linear: $G = 10^{G_{dB}/10}$

Step 2: Calculate wavelength

$$\lambda_0 = \frac{c}{f}$$

Step 3: Calculate power density at distance r

$$P_d^{inc} = \frac{P_{trans} \cdot G_{trans}}{4\pi r^2}$$

Units: W/m²

Step 4: Calculate effective aperture of receive antenna

$$A_{eff} = \frac{\lambda_0^2}{4\pi} G_{rec}$$

For dish antenna with aperture efficiency e_{ap} :

$$A_{eff} = e_{ap} \cdot \pi \left(\frac{D}{2} \right)^2$$

Step 5: Calculate received power

Method 1:

$$P_{rec} = P_d^{inc} \cdot A_{eff}$$

Method 2 (Friis directly):

$$P_{rec} = P_{trans} \cdot G_{trans} \cdot G_{rec} \cdot \left(\frac{\lambda_0}{4\pi r} \right)^2$$

Step 6: Calculate Thévenin voltage (if asked)

For conjugate-matched load ($R_L = R_{ant}$):

$$P_{rec} = \frac{|V_{Th}|^2}{8R_{ant}}$$

$$|V_{Th}| = \sqrt{8R_{ant} \cdot P_{rec}}$$

Step 7: Convert to dBm or dBW (if asked)

$$P_{dBm} = 10 \log_{10} \left(\frac{P_{watts}}{0.001} \right)$$

$$P_{dBW} = 10 \log_{10}(P_{watts})$$

WARNING: The $\left(\frac{\lambda_0}{4\pi r}\right)^2$ term is NOT the “path loss” — it includes the receive antenna’s aperture. Don’t double-count!

WARNING: Common error: forgetting to square the $(\lambda/4\pi r)$ term.

TIP: Received power is typically TINY (pW to fW range). This is normal for satellite links!

8.3 EIRP (Effective Isotropic Radiated Power)

EIRP is the power that would be needed from an isotropic antenna to produce the same power density:

$$EIRP = P_{trans} \cdot G_{trans}$$

In dB:

$$EIRP_{dBW} = P_{trans,dBW} + G_{trans,dB}$$

8.4 Free-Space Path Loss (FSPL)

The “spreading loss” as wave travels distance r :

$$FSPL = \left(\frac{4\pi r}{\lambda_0} \right)^2 = \left(\frac{4\pi r f}{c} \right)^2$$

In dB:

$$FSPL_{dB} = 20 \log_{10} \left(\frac{4\pi r}{\lambda_0} \right) = 20 \log_{10}(r) + 20 \log_{10}(f) + 20 \log_{10} \left(\frac{4\pi}{c} \right)$$

Simplified (with r in km, f in GHz):

$$FSPL_{dB} \approx 92.45 + 20 \log_{10}(r_{km}) + 20 \log_{10}(f_{GHz})$$

8.5 Link Budget in dB Form

$$P_{rec,dBm} = P_{trans,dBm} + G_{trans,dB} - FSPL_{dB} + G_{rec,dB}$$

Or equivalently:

$$P_{rec,dBm} = EIRP_{dBm} - FSPL_{dB} + G_{rec,dB}$$

8.6 Polarization Mismatch Factor

If transmit and receive antennas have different polarizations:

$$P_{rec} = P_{rec,matched} \cdot |\hat{\rho}_{rec} \cdot \hat{\rho}_{trans}|^2$$

Polarization Match	Mismatch Factor
Same linear polarization	1 (0 dB loss)
Orthogonal linear	0 (complete loss)
Linear to circular	0.5 (−3 dB)
RHCP to LHCP	0 (complete loss)
Same circular (RHCP-RHCP)	1 (0 dB loss)

8.7 Directivity vs. Gain

Directivity D : Ratio of max radiation intensity to average (theoretical, no losses)

Gain G : Includes antenna efficiency e_r (ohmic losses, mismatch)

$$G = e_r \cdot D$$

For lossless antenna: $G = D$

Half-Power Beamwidth (HPBW) Approximation:

$$D \approx \frac{4\pi}{\Omega_A} \approx \frac{41,253}{\theta_{1,deg} \cdot \theta_{2,deg}}$$

where θ_1 and θ_2 are the HPBW in the two principal planes (in degrees).

8.8 Radar Range Equation (Bonus)

For monostatic radar (same antenna transmits and receives):

$$P_{rec} = \frac{P_{trans} G^2 \lambda_0^2 \sigma}{(4\pi)^3 r^4}$$

where σ is the radar cross-section of the target [m²].

9 TDR and Bounce Diagrams

9.1 Problem Type A: Construct Bounce Diagram

Step 1: Calculate reflection coefficients

$$\Gamma_L = \frac{R_L - Z_0}{R_L + Z_0} \quad \Gamma_g = \frac{R_g - Z_0}{R_g + Z_0}$$

Step 2: Calculate initial forward wave amplitude

$$V^+ = V_0 \cdot \frac{Z_0}{R_g + Z_0}$$

Step 3: Calculate one-way travel time

$$T = \frac{L}{v_p} = \frac{L\sqrt{\epsilon_r}}{c}$$

Step 4: Draw the bounce diagram

- Horizontal axis: position ($z = 0$ at generator, $z = L$ at load)
- Vertical axis: time (increasing downward)
- Draw diagonal lines for waves bouncing back and forth

Step 5: Label wave amplitudes

- First wave: V^+
- After load reflection: $\Gamma_L V^+$
- After generator reflection: $\Gamma_g \Gamma_L V^+$
- Continue pattern: multiply by Γ_L or Γ_g at each bounce

Step 6: Find voltage at any point

Sum all waves that have passed that point by the given time.

9.2 Problem Type B: Interpret TDR Measurement

Given: Voltage vs. time plot at $z = 0$. Find line length and load resistance.

Step 1: Find initial voltage V_1

$$V_1 = V^+ = V_0 \cdot \frac{Z_0}{R_g + Z_0}$$

Step 2: Identify round-trip time t_{rt}

This is when the voltage changes (reflected wave returns).

Step 3: Calculate line length

$$L = \frac{v_p \cdot t_{rt}}{2} = \frac{c \cdot t_{rt}}{2\sqrt{\epsilon_r}}$$

Step 4: Use voltage change to find Γ_L

At $t = t_{rt}$, voltage becomes:

$$V_2 = V^+(1 + \Gamma_L + \Gamma_L \Gamma_g)$$

Solve for Γ_L :

$$\Gamma_L = \frac{V_2/V^+ - 1}{1 + \Gamma_g}$$

Step 5: Calculate load resistance

$$R_L = Z_0 \cdot \frac{1 + \Gamma_L}{1 - \Gamma_L}$$

9.3 Problem Type C: Pulse Response

Step 1: Treat pulse as step minus delayed step

Pulse of width W starting at $t = 0$:

$$v_{pulse}(t) = V_0[u(t) - u(t - W)]$$

Step 2: Draw bounce diagram for step function

Step 3: At observation point, voltage due to pulse

For each wave arrival time t_i :

- Wave turns “on” at t_i
- Wave turns “off” at $t_i + W$

Step 4: Sum contributions in each time interval

WARNING: For pulse problems, each wave creates TWO voltage transitions: rising edge and falling edge (delayed by W).

TIP: The final DC voltage (after all bounces settle) must equal:

$$V_{final} = V_0 \cdot \frac{R_L}{R_g + R_L}$$

Use this to check your answer!

9.4 Problem Type D: Reactive Loads (Capacitor or Inductor)

Reactive loads create TIME-VARYING reflection coefficients.

Capacitor Load:

- At $t = 0^+$: Capacitor is short circuit $\rightarrow \Gamma_L = -1$
- At $t = \infty$: Capacitor is open circuit $\rightarrow \Gamma_L = +1$
- Time constant: $\tau = Z_0 C$

$$\Gamma_L(t) = 1 - 2e^{-t/\tau} \quad (\text{starts at } -1, \text{ ends at } +1)$$

Inductor Load:

- At $t = 0^+$: Inductor is open circuit $\rightarrow \Gamma_L = +1$
- At $t = \infty$: Inductor is short circuit $\rightarrow \Gamma_L = -1$
- Time constant: $\tau = L/Z_0$

$$\Gamma_L(t) = -1 + 2e^{-t/\tau} \quad (\text{starts at } +1, \text{ ends at } -1)$$

Voltage at Load:

$$V_L(t) = V^+(1 + \Gamma_L(t))$$

TIP: Memory trick: Capacitor “looks like” a short at high frequency (fast changes), open at DC. Inductor is opposite.

9.5 Problem Type E: Multiple Discontinuities

If there are multiple impedance changes along the line:

Step 1: Calculate reflection coefficient at each discontinuity

At junction between Z_{01} and Z_{02} :

$$\Gamma_{12} = \frac{Z_{02} - Z_{01}}{Z_{02} + Z_{01}}$$

Step 2: Calculate transmission coefficient

$$T_{12} = 1 + \Gamma_{12} = \frac{2Z_{02}}{Z_{02} + Z_{01}}$$

Step 3: Track waves through each section

- Wave entering section: multiply by T
- Wave reflecting: multiply by Γ
- Wave passing through: picks up phase delay $e^{-j\beta l}$ (or time delay l/v_p)

Step 4: Sum all contributions at point of interest

9.6 Input Impedance with Multiple Sections

For cascaded transmission line sections, work from load backward:

Step 1: Start with Z_L at the end

Step 2: Transform through last section: $Z_{in,N} = Z_{0N} \frac{Z_L + jZ_{0N} \tan \beta_N l_N}{Z_{0N} + jZ_L \tan \beta_N l_N}$

Step 3: Use $Z_{in,N}$ as the “load” for the next section

Step 4: Repeat until you reach the input

10 Bonus: Poynting Vector and Power Flow

10.1 Poynting Vector

The Poynting vector gives the power flow per unit area:

$$\vec{S} = \vec{E} \times \vec{H} \quad [\text{W/m}^2]$$

Time-Average Poynting Vector (for sinusoidal fields):

$$\langle \vec{S} \rangle = \frac{1}{2} \text{Re}\{\vec{E} \times \vec{H}^*\}$$

10.2 Power Flow in Transmission Lines

Forward Wave Power:

$$P^+ = \frac{|V^+|^2}{Z_0} = |I^+|^2 Z_0$$

Reflected Wave Power:

$$P^- = |\Gamma|^2 P^+$$

Net Power Delivered to Load:

$$P_L = P^+ - P^- = P^+(1 - |\Gamma|^2)$$

10.3 Power in Plane Waves

Power Density:

$$\langle S \rangle = \frac{|E_0|^2}{2\eta} = \frac{1}{2}\eta|H_0|^2 \quad [\text{W/m}^2]$$

For lossless medium:

$$\eta = \sqrt{\frac{\mu}{\epsilon}} = \frac{\eta_0}{\sqrt{\epsilon_r}} = \frac{377}{\sqrt{\epsilon_r}} \quad [\Omega]$$

10.4 Polarization Review

Linear Polarization: E-field oscillates in one direction

- Horizontal (TE): E-field parallel to ground
- Vertical (TM): E-field perpendicular to ground

Circular Polarization: E-field rotates, constant magnitude

- RHCP: Rotates clockwise when looking toward source
- LHCP: Rotates counter-clockwise when looking toward source
- Requires two orthogonal components with 90° phase difference and equal magnitude

Elliptical Polarization: E-field traces ellipse (general case)

$$AR = \frac{E_{major}}{E_{minor}} \geq 1$$

- $AR = 1$: Circular polarization
- $AR = \infty$: Linear polarization

You got this, I love you :3